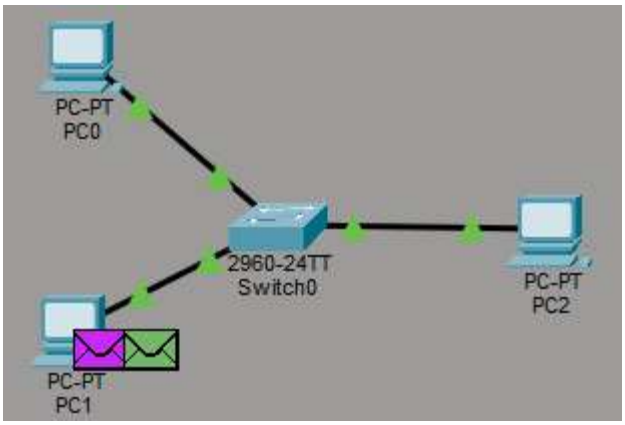
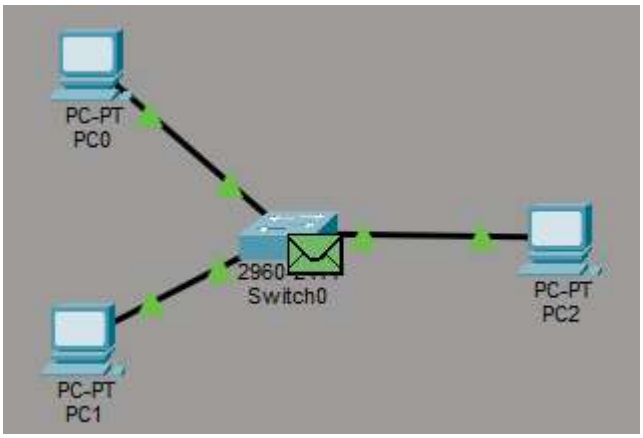
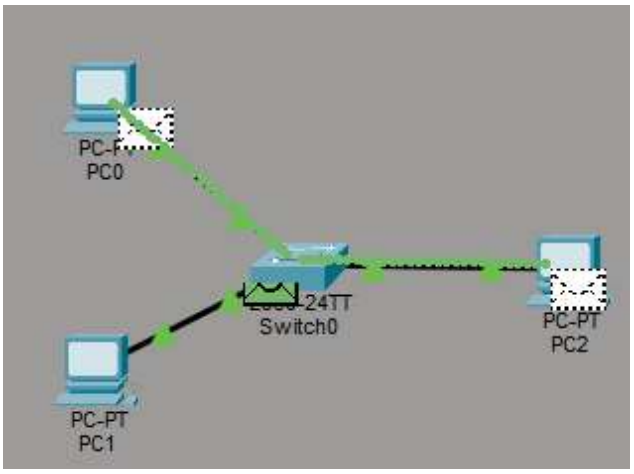
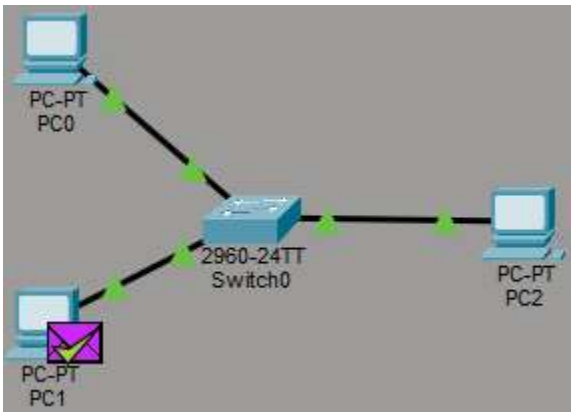
1)


PC-PT
PC0
2960-24TT
Switch0
PC-PT
PC2
PC-PT
PC1


PC-PT
PC0
Switch0
PC-PT
PC2
PC-PT
PC1


PC0
Switch0
PC-PT
PC2
PC-PT
PC1

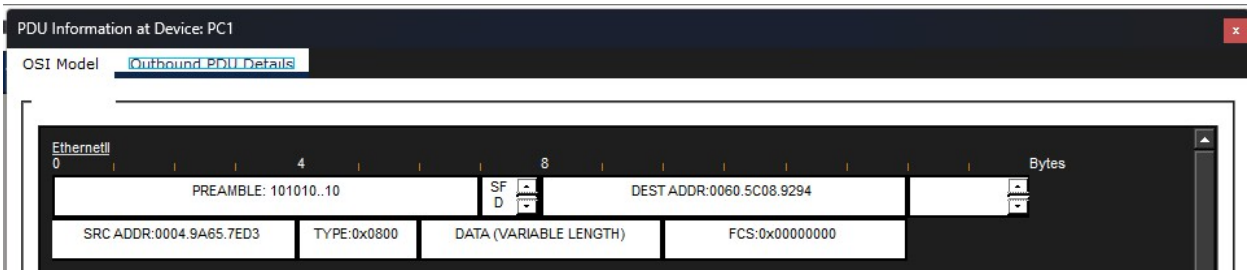



```
Switch#enable
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.1718.e4d3    DYNAMIC     Fa0/1
   1    0004.9a65.7ed3    DYNAMIC     Fa0/2
   1    0060.5c08.9294    DYNAMIC     Fa0/3
Switch#
```

2)

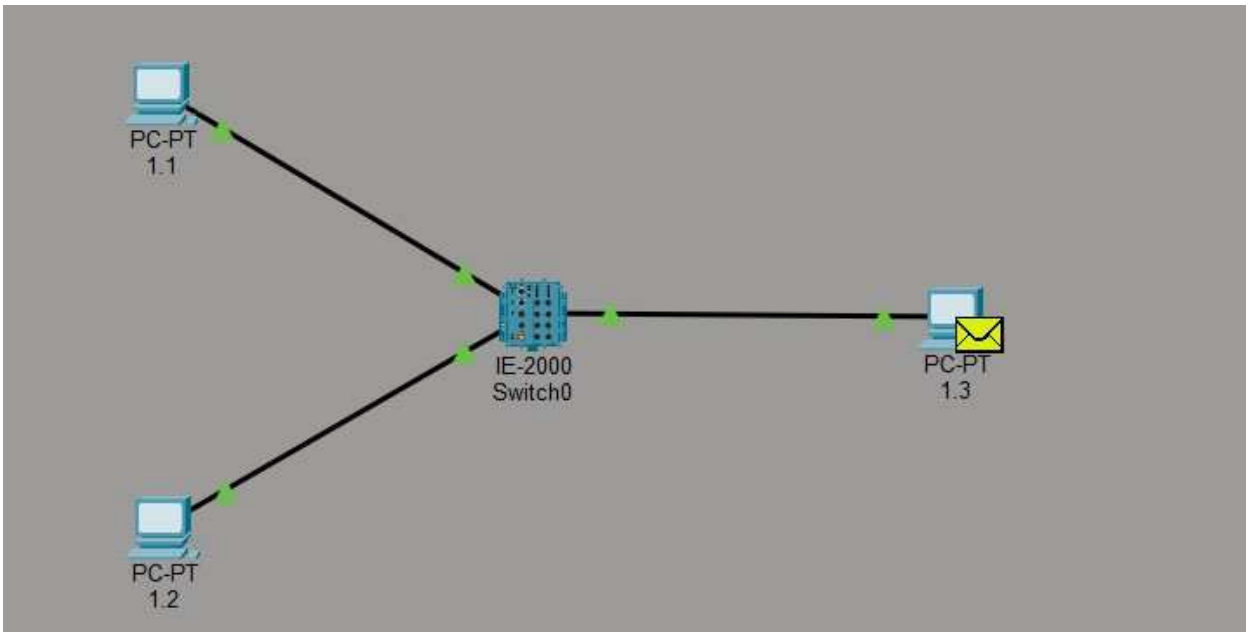


3)PING

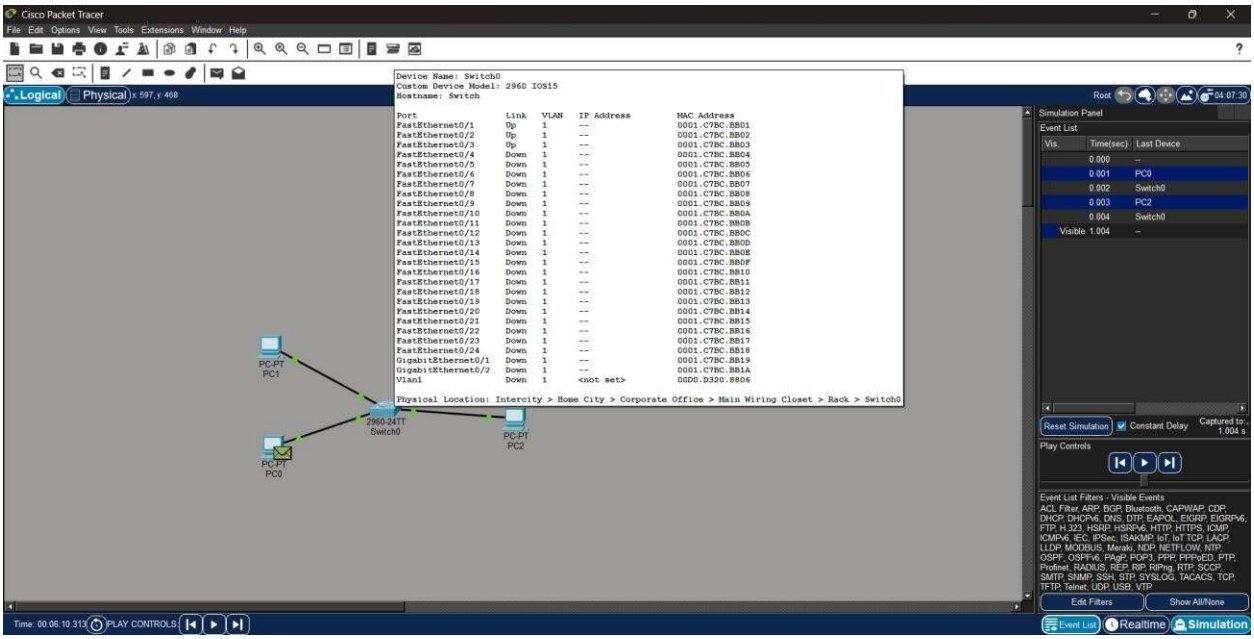
```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0009.7c39.80c2    DYNAMIC     Fa1/3
   1    00d0.bc42.e736    DYNAMIC     Fa1/1
Switch#
```

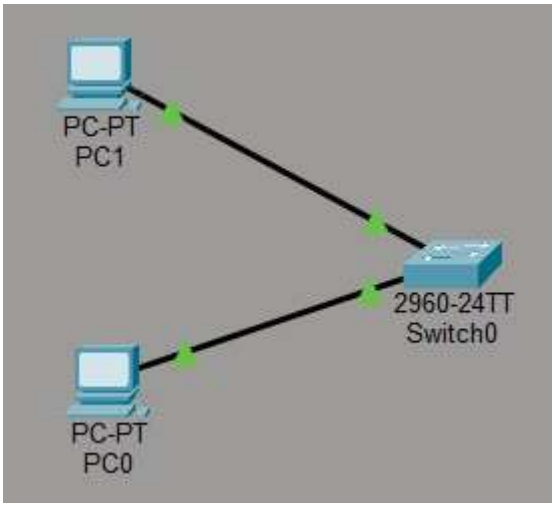


AFTER CHANGING MAC ADDR



4)

LAN SETUP



PING

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=8ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 8ms, Average = 5ms

LINUX KERNAL HANDLING OF ETHERNET DEVICES

The Linux kernel handles Ethernet communication through a structured networking stack.

Key Components:

- **NIC Driver**: Controls hardware (e.g., Ethernet card)
- **net_device structure**: Abstract representation of network interface

- **Socket Buffer (sk_buff)**: Data structure for packet handling

Packet Flow:

1. Packet received by NIC
2. Interrupt triggered → handled by driver
3. Packet stored in **sk_buff**
4. Passed through layers:
    a. Data Link → Network → Transport
5. Delivered to application

Mac Address Table of a Switch

```
Switch>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.64be.d316    DYNAMIC     Fa0/1
   1    00e0.f98c.1b12    DYNAMIC     Fa0/2
Switch>
```

1. Linux Kernel Handling of Ethernet Devices

The Linux kernel manages Ethernet communication through a layered networking architecture integrating user space, protocol stacks, device drivers, and hardware interfaces.

(a) Network Stack Layers

User applications interact via the socket API, which interfaces with the kernel TCP/IP stack, followed by NIC drivers that communicate with the physical Ethernet hardware.

(b) Core Kernel Modules

The net/core/ module handles packet processing, net/ipv4/ and net/ipv6/ implement protocol logic, and drivers/net/ethernet/ contains hardware-specific NIC drivers.

(c) Data Flow (Packet Path)

During transmission, data flows from application to socket, through transport and IP layers, then encapsulated into Ethernet frames and sent via the NIC driver.

During reception, frames are captured by the NIC, processed via interrupts or NAPI, passed up through protocol layers, and delivered to the application.

(d) Key Kernel Mechanisms

NAPI optimizes packet processing by reducing interrupts, SKB structures manage packet data,

Netfilter enables packet filtering, and the bridge module provides Layer 2 switching functionality.

2. Configuring a Basic LAN Interface using ip

The ip utility provides a modern interface for configuring and managing network devices in Linux.

(a) View Interfaces

The ip link show command displays all available network interfaces and their states.

(b) Enable Interface

The command sudo ip link set eth0 up activates the specified network interface.

(c) Assign IP Address

The command sudo ip addr add 192.168.1.10/24 dev eth0 assigns an IPv4 address to the interface.

(d) Configure Default Gateway

The command sudo ip route add default via 192.168.1.1 sets the default route for outbound traffic.

(e) Verify Configuration

The commands ip addr show eth0 and ip route confirm interface and routing configurations.

(f) Test Connectivity

The ping 192.168.1.1 command verifies network reachability.

3. Viewing MAC Address Table using Linux Bridge

Linux implements software-based switching through bridge interfaces that operate at Layer 2.

(a) Create Bridge Interface

The commands sudo ip link add br0 type bridge and sudo ip link set br0 up create and enable a bridge.

(b) Add Interfaces to Bridge

The commands sudo ip link set eth0 master br0 and sudo ip link set eth1 master br0 attach interfaces to the bridge.

4. Inspecting the MAC Address Table

The bridge fdb show command displays the forwarding database maintained by the bridge.

Explanation

Each entry maps a learned MAC address to a specific interface, indicating where frames should be

forwarded.

Dynamic entries are learned through traffic, while static entries can be manually configured.

Conclusion:

Ethernet communication in a LAN is achieved through ARP-based address resolution and MAC-based frame forwarding, while the Linux kernel efficiently manages packet processing through its networking stack.P